Increasing exponential functions

1 Consider the function $y = 3^x$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behaviour of the function as x increases.

c State the domain of the function.

d State the range of the function.

2 Consider the expression 2^x.

a Evaluate the expression when $x = -4$.

b Evaluate the expression when $x = 0$.

c Evaluate the expression when $x = 4$.

d For what values of x will 2^x have a negative value?

e Describe what happens to the value of 2^x as x gets larger.

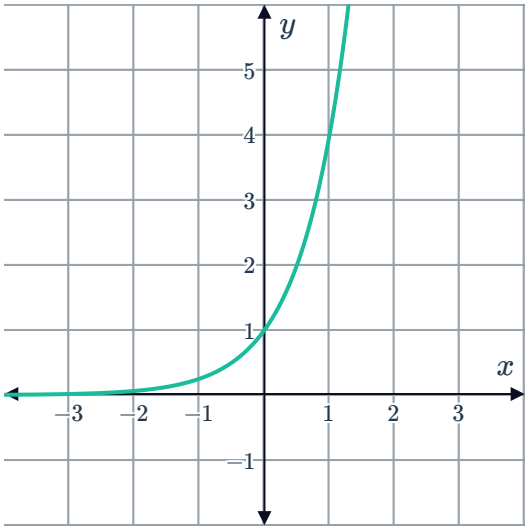
f Describe what happens to the value of 2^x as x gets smaller.

3 Consider the graph of the equation $y = 4^x$:

a Is each y-value of the function positive or negative?

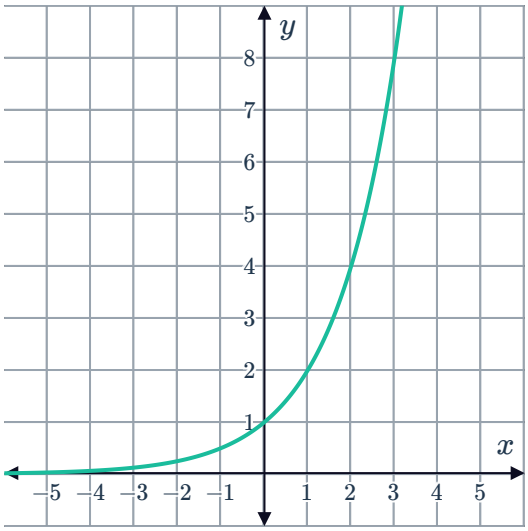
b State the value of y the graph approaches but does not reach.

c State the equation and name of the horizontal line, which $y = 4^x$ gets closer and closer to but never intersects.



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4 The graph of $y = 2^x$ is shown:



a State the y-intercept of this graph.

b Does the graph have an x-intercept?

c State the domain of the function.

d State the range of the function.

e Find the value of y when $x = 7$.

f Find the value of x when $y = 256$.

5 Consider the function $y = 6^x$.

a Can the value of y ever be zero or negative? Explain your answer.

b As the values of x get larger and larger, what value does y approach?

c As the values of x get smaller and smaller, what value does y approach?

d Find the y-value of the y-intercept of the curve.

e How many x-intercepts does the curve have?

f Sketch the graph of $y = 6^x$.

6 Consider the functions $y = 2^x$, $y = 3^x$ and $y = 5^x$.

a Sketch the three functions on the same set of axes.

b State the y-intercept of each curve.

c Describe the nature of these functions for increasingly large values of x.

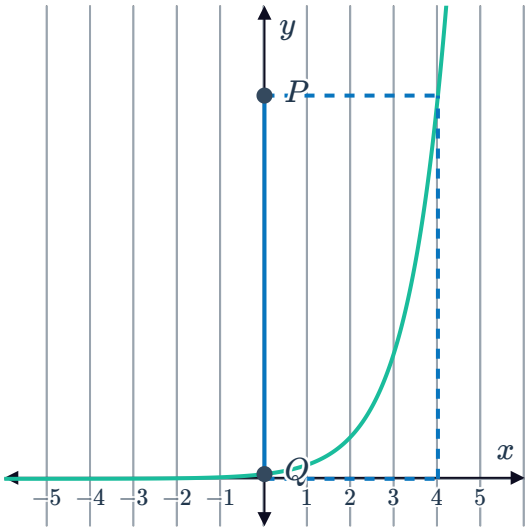
7 If the point $(3, m)$ lies on the curve $y = 2^x$, solve for m.

8 Find the missing coordinate in each ordered pair so that the pair is a solution to $y = 5^x$:

a $(3, \square)$ **b** $(\square, 5)$ **c** $(-1, \square)$ **d** $\left(\square, \frac{1}{25}\right)$

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9 Consider the graph of $y = 3^x$. Find the length of the interval PQ.



Decreasing exponential functions

10 Consider the function $y = 2^{-x}$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behaviour of the function as x increases.

c Find the y-value of the y-intercept of the function.

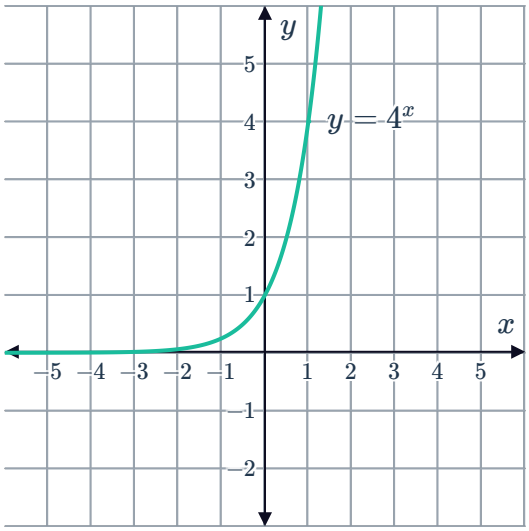
d State the domain of the function.

e State the range of the function.

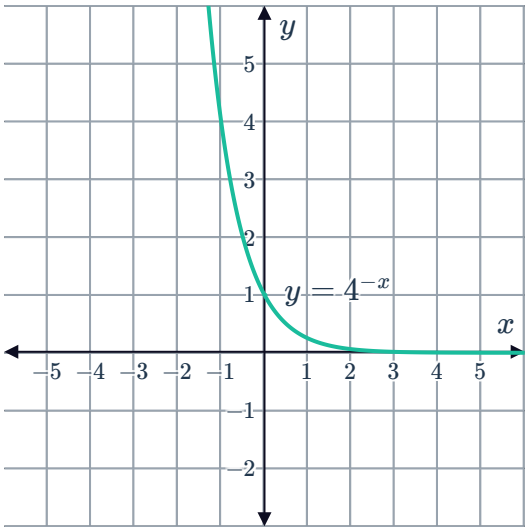
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11 Consider the graphs of the functions $y = 4^x$ and $y = 4^{-x}$ below. Describe the rate of change for each function.

a



b



12 Consider the function $y = 3^{-x}$.

a Find the y-value of the y-intercept of the curve.

b Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

c State the equation of the horizontal asymptote of the curve.

d Sketch the curve $y = 3^{-x}$.

13 Consider the functions $y = 2^{-x}$, $y = 3^{-x}$ and $y = 5^{-x}$.

a Sketch the three functions on the same set of axes.

b State the y-intercept of each curve.

c Describe the nature of these functions for large values of x.

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14 Consider the function $y = 8^{-x}$.

a Can the value of y ever be zero or negative? Explain your answer.

b As the value of x gets larger and larger, what value does y approach?

c As the value of x gets smaller and smaller, what value does y approach?

d Find the y-value of the y-intercept of the curve.

e How many x-intercepts does the curve have?

f Sketch the graph of $y = 8^{-x}$.

15 Consider the function $y = \left(\frac{1}{2}\right)^x$.

a Determine whether the following functions are equivalent to $y = \left(\frac{1}{2}\right)^x$:

i $y = \frac{1}{2^x}$ **ii** $y = 2^{-x}$ **iii** $y = -2^x$ **iv** $y = -2^{-x}$

b Hence, describe a transformation that would obtain the graph of $y = \left(\frac{1}{2}\right)^x$ from the graph of $y = 2^x$.

c Graph the functions $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ on the same set of axes.

16 Consider the functions $y = 5^x$ and $y = \left(\frac{1}{5}\right)^{-x}$.

a Sketch the graph of $y = 5^x$.

b Complete the table of values for $y = \left(\frac{1}{5}\right)^{-x}$:

x	-3	-2	-1	0	1	2	3
y							

c Sketch the graph of $y = \left(\frac{1}{5}\right)^{-x}$.

d Compare the graphs of $y = 5^x$ and $y = \left(\frac{1}{5}\right)^{-x}$.

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17 Consider the function $y = \left(\frac{1}{3}\right)^x$.

a Rewrite the function in the form $y = k^{-x}$.

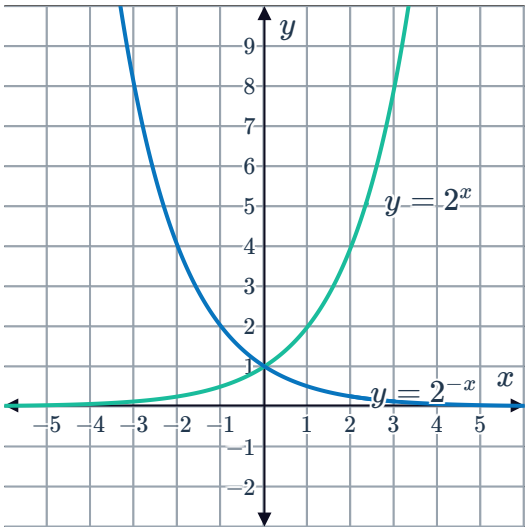
b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{3}\right)^x$ from the graph of $y = 3^x$.

c Graph the functions $y = 3^x$ and $y = \left(\frac{1}{3}\right)^x$ on the same set of axes.

18 Consider the graph of the following functions $y = 2^x$ and $y = 2^{-x}$:

a State the coordinates of the point of intersection of the two curves.

b Describe the behaviour of both these functions for large values of x.



19 Determine whether following are increasing or decreasing exponential functions:

a $y = 3^x$ **b** $y = \left(\frac{3}{5}\right)^x$ **c** $y = 0.5^x$ **d** $y = 1.05^x$

e $y = \left(\frac{4}{3}\right)^x$ **f** $y = 0.97^x$ **g** $y = 1.5^{-x}$ **h** $y = \left(\frac{1}{4}\right)^{-x}$

20 Do either of the functions $y = 9^x$ or $y = 9^{-x}$ have x-intercepts? Explain your answer.

21 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = 3^{-x}$:

a $(0, \square)$ **b** $(\square, 9)$ **c** $(-1, \square)$ **d** $\left(\square, \frac{1}{27}\right)$

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Evaluate exponential functions

22 Consider the function $y = 5 \times 4^x$. Find the value of y when:

a $x = 0$ **b** $x = 2$ **c** $x = -1$

23 Consider the function $f(x) = 3 + 4^{x+5}$. Evaluate $f(-2)$.

24 Consider the function $f(x) = 2 + 3^x$.

a Evaluate $f(4)$. **b** Evaluate $f(-3)$.

25 Consider the function $f(x) = \left(\frac{4}{3}\right)^x$.

a Evaluate $f(0)$. **b** Evaluate $f(-2)$.

26 Consider the function $f(x) = 2^{x-3}$.

a Evaluate $f(0)$.

b Evaluate $f\left(\frac{1}{4}\right)$. Give your answer in exact positive index form.

27 Consider the function $f(x) = 3^x + 3^{-x}$.

a Evaluate $f(3)$. **b** Evaluate $f(-3)$.

c Is $f(3) = f(-3)$?

28 If $f(x) = 5^x$ and $g(x) = 3^{-x}$, evaluate:

a $f(1)$ **b** $g(f(1))$ **c** $f(g(0))$

29 Find the value of $5000(1.08^x)$ for $x = 30$ to two decimal places.

30 Find the value of $4000(0.02)^{0.6^t}$ for $t = 4$ to two decimal places.

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Compare growth

31 Consider the functions $f(x) = 3x$ and $g(x) = 3^x$.

a Complete the table of values for each function:

x	0	1	2	3	4
$f(x)$					
$g(x)$					

b Graph the functions on the same set of axes.

c

i For every 1 unit increase in x, the y value of $f(x)$ increases by how many units?

ii For every 1 unit increase in x, the y value of $g(x)$ increases by a factor of what?

d For $x \geq 1$, which function increases at a faster rate?

e Find the following function values:

i $f(8)$ **ii** $g(8)$

f Describe the behaviour of each function as x increases.

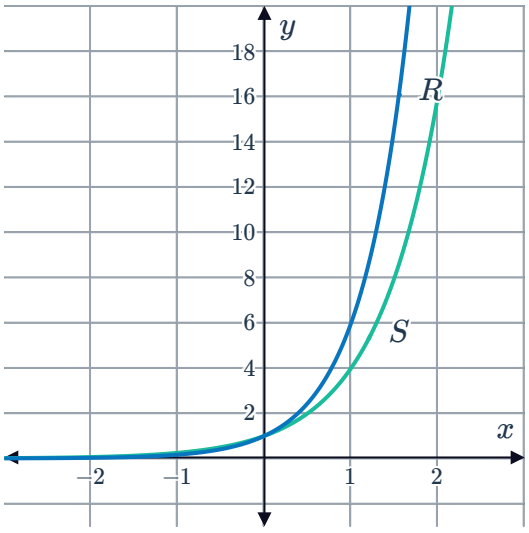
32 Consider the two functions $y = 4^x$ and $y = 5^x$. Which function increases more rapidly for $x > 0$?

33 Consider the two functions $y = 4^{-x}$ and $y = 5^{-x}$. Which function decreases more rapidly for $x > 0$?

34 Consider the graphs of the two exponential functions R and S:

a One of the graphs is of $y = 4^x$ and the other graph is of $y = 6^x$.
Which is the graph of $y = 6^x$?

b For $x < 0$, is the graph of $y = 6^x$ above or below the graph of $y = 4^x$. Explain your answer.



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35 Consider the table of values for the functions for $x \geq 1$:
$f(x) = (1.05)^x$ and $g(x) = 5x$.
Is Ivan correct to conclude that $g(x)$ is always greater than $f(x)$ for all values of $x \geq 1$. Explain your answer.

x	1	2	3	4	5
$f(x)$	1.05	1.10	1.16	1.22	1.28
$g(x)$	5	10	15	20	25

36 Two different sequences are generated by the functions $f(n) = 5n + 1$ and $g(n) = 3(2)^n$.

a Find the first 5 terms in the sequence generated by each function.

n	1	2	3	4	5
$f(n)$					
$g(n)$					

b Describe how the terms of each sequence increase.

c Is John correct to conclude that the terms of $g(n)$ will always be greater than the terms of $f(n)$ for $n > 1$? Why?

37 Consider the linear function $f(x) = 5x + 2$ and the exponential function $g(x) = 5(3)^x$.

a Complete the table of values:

x	4	5	6
$f(x)$			
$g(x)$			

b Evaluate:

i $f(5) - f(4)$

ii $\dfrac{g(5)}{g(4)}$

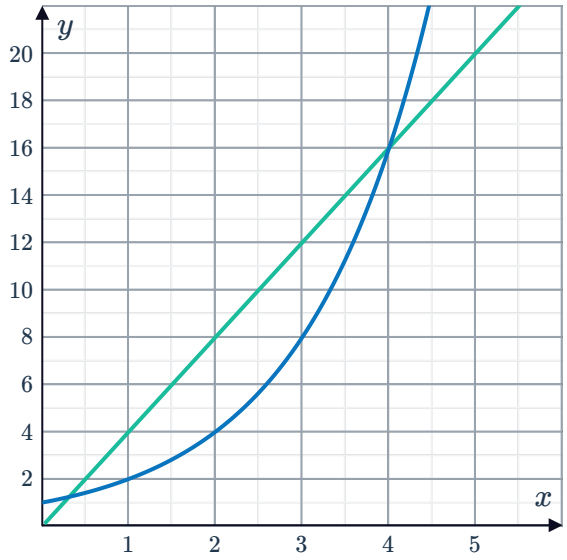
iii $f(k+1) - f(k)$

iv $\dfrac{g(k+1)}{g(k)}$

c For each 1 unit increase in x, describe how the linear and exponential functions increase.

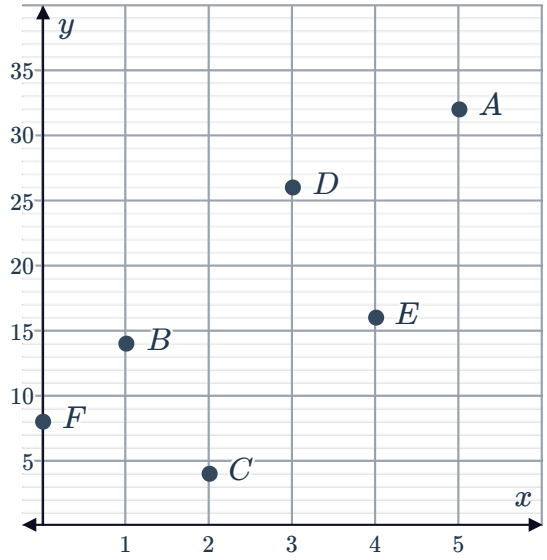
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38 A linear function and exponential function have been graphed on the same axes:



- **a** For each 1 unit increase in x, by how much does the linear function increase?
- **b** For each 1 unit increase in x, by what multiplicative factor does the exponential function increase?
- **c** As x approaches infinity, which function increases more rapidly?

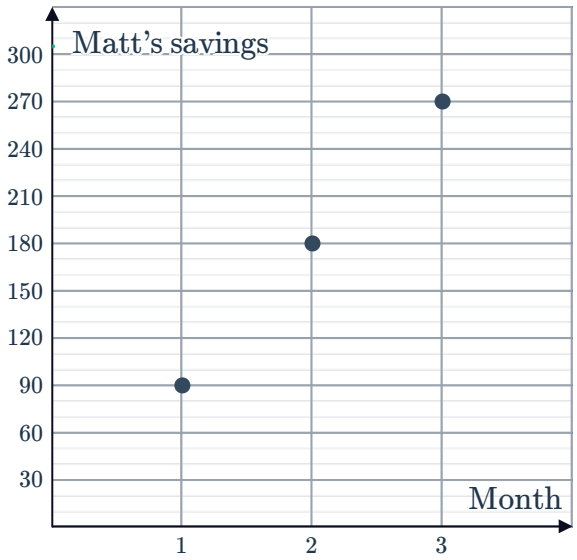
39 Several points have been plotted on the number plane:



- **a** Find the three points that form a linear relationship between x and y.
- **b** Find the three points that form an exponential relationship between x and y.
- **c** For each 1 unit increase in x, by how much does the linear function increase?
- **d** For each 1 unit increase in x, by what multiplicative factor does the exponential function increase?

40 Matt and Sophia are saving money using different strategies. The amount each has saved after each month is given by the table of values and the plotted points. If each person continues their pattern of saving, who will be the first to exceed savings of \$600?

Number of months	1	2	3
Sophia's savings	4	16	64



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41 Consider the functions $f(x) = 2^x$ and $g(x) = 2x^2$ for $x \geq 0$.

a Complete the table of values for each function:

x	$f(x)$	Increase in $f(x)$	$g(x)$	Increase in $g(x)$
0		–		–
1		1		2
2		2		6
3				
4				
5				
6				
7				

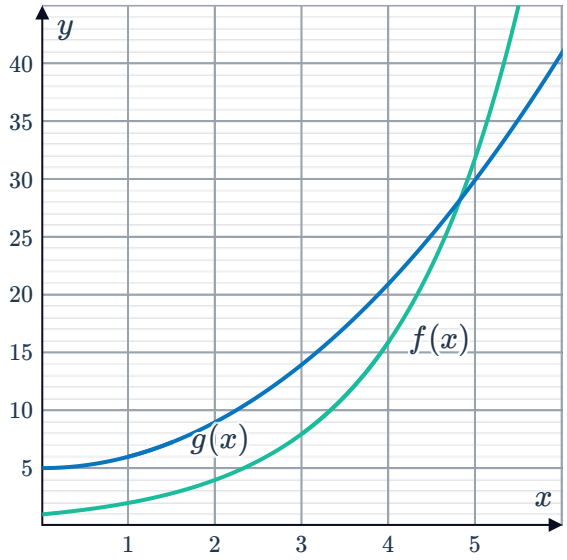
b Describe the behaviour of the two functions.

42 Kate considers the given graphs of $f(x)$ and $g(x)$ and performs the following calculations:

- Calculation 1: $f(11) - f(10) = 1024$
- Calculation 2: $f(10) - g(10) = 919$
- Calculation 3: $f(11) - g(11) = 1922$
- Calculation 4: $g(11) - g(10) = 21$

She claims that for large values of x, as x increases, the exponential function increases more rapidly than the quadratic function.
Explain how the following pairs of calculations support the claim:

a 1 and 4. **b** 2 and 3.



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